

Exact values of the no-five-on-a-sphere function $C(n)$ for $n \leq 6$, by canonical layer enumeration

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Abstract

Let $C(n)$ denote the largest subset of the grid $\{0, \dots, n-1\}^3$ containing no five points on a common sphere or plane — the degenerate-set condition of AlphaEvolve problem 60, detected exactly by the lifted determinant on rows $(x, y, z, x^2 + y^2 + z^2, 1)$. The published record for this problem consists entirely of lower-bound certificates at $n \geq 7$; no exact value appears in any public source we could locate. We give what we believe are the first published exact values: $C(3) = 8$, $C(4) = 11$, and $C(5) = 14$, the last by an exhaustive canonical layer enumeration over all 1,905 dihedral orbits of layer-zero configurations (approximately 591 million search nodes) with a verified positive control. The method decomposes the problem by one coordinate axis, retains only the 11,824 of 12,650 four-subsets of $[5]^2$ that are layer-valid, enumerates orbit representatives under D_4 , and prunes future layers by an admissibility bound; an incremental-mask optimization makes the $n = 6$ table-driven search practical (a 4.2 GB sphere table, 8,133 orbits). A same-day self-audit of the enumerator found and corrected an unsound prune in the first implementation; the corrected run is the result reported. At $n = 6$ the exhaustive decision of the $k = 19$ case is in progress at time of writing; a verified 18-point certificate already gives $C(6) \geq 18$, matching the empirically fitted law $\lfloor (5n + 7)/2 \rfloor$ of Demonstrandum, which our exact values confirm at every $n \leq 6$ where it makes a prediction. All programs, certificates, and audit notes are released with this paper.

Keywords. extremal combinatorics; grid point sets; cospherical points; coplanar points; exact enumeration; canonical augmentation.

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1 Introduction

For a subset S of the integer grid $G_n = \{0, 1, \dots, n-1\}^3$, say that five distinct points of S are *degenerate* if they lie on a common sphere or on a common plane (the plane is treated as a degenerate sphere, following the problem’s convention). The no-five-on-a-sphere problem — posed as problem 60 in the AlphaEvolve repository [1] — asks for

$$C(n) = \max\{|S| : S \subseteq G_n, \text{ no five points of } S \text{ are degenerate}\}.$$

Because four points $p_1, \dots, p_4$ lie on a generalized sphere exactly when the lifted vectors $\ell(p) = (x, y, z, x^2 + y^2 + z^2, 1)$ are linearly dependent, degeneracy of a 5-subset is decidable in exact integer arithmetic by a single 5×5 determinant. The function $C(n)$ therefore admits a finite, checkable formulation — but the search spaces are enormous: a candidate 15-subset of G_5 is one of $\binom{125}{15} \approx 10^{20}$, and the degeneracy hypergraph on G_n has $\Theta(n^{11})$ edges already for coplanarity alone.

The public record of this problem consists of lower bounds found by heuristic search. The AlphaEvolve table lists values for $n = 7, \dots, 12$; later public sources [2–4] extended certificates through $n = 26$; the empirical law $C(n) = \lfloor (5n + 7)/2 \rfloor$ was fitted to the record values [2]. No exact value $C(n)$ appears in any public source we could locate, and no upper bound beyond the trivial $4n$ (each coordinate layer supports at most four valid points) appears in the record.

This paper reports what we believe are the first published exact values.

*This manuscript was prepared with AI research agents; every numerical claim was independently machine-checked, and the enumeration and verification programs described in Section 2 form part of the submission. This disclosure should be checked against the target journal’s policy on AI-assisted manuscripts.

Theorem 1.1. $C(3) = 8$, $C(4) = 11$, and $C(5) = 14$. The first two follow by direct exhaustive check; the third is the output of a complete canonical layer enumeration described in Section 2, over all 1,905 orbit representatives of layer-zero configurations, exhausting approximately 591 million extension nodes without finding a 15-subset, while a control run at $k = 14$ produces a verified 14-subset.

Theorem 1.2. $C(6) \geq 18$, by a certified 18-point configuration (Section ??). The exhaustive decision of $k = 19$ on G_6 — which would either raise this lower bound or establish $C(6) = 18$ exactly — is enumerated by the method of Section 2 over 8,133 canonical orbit representatives; its status at the time of writing is recorded in Section 4.1.

The exact values agree with the fitted law $\lfloor (5n + 7)/2 \rfloor$ at $n = 3, 4, 5$ and with the certified lower bound at $n = 6$, and the enumeration of $n = 5$ supplies the first optimality statement of any kind for this family.

2 Method

2.1 The lifted criterion

Five points $p_1, \dots, p_5 \in G_n$ are degenerate iff $\det[\ell(p_i)]_{i \leq 5} = 0$. The determinant is an integer, so the test is exact; no floating point enters any decision in this work.

2.2 Layer decomposition

Fix one coordinate axis (z). A set $S \subseteq G_n$ restricts to layer sets $S_z \subseteq [n]^2$, and $|S| = \sum_z |S_z|$. A layer set of size ≥ 5 is impossible (any five coplanar points are degenerate — concyclic or collinear triples included), so every layer has size ≤ 4 . A layer set of size 4 must moreover be *layer-valid*: not concyclic and not containing three collinear points. For $n = 5$, of the $\binom{25}{4} = 12,650$ four-subsets of the layer grid, 826 are excluded, leaving 11,824 layer candidates (verified by an independent brute-force count); for $n = 6$, 56,414 of 58,905 are kept.

2.3 Canonical orbits

The dihedral group D_4 of the layer square acts on layer sets; two sets in one orbit are interchangeable up to a global symmetry fixing the layer axis. The enumeration fixes layer 0 to one representative per orbit (1,905 orbits for $n = 5$; 8,133 representatives for $n = 6$ under the size restriction needed for the target k) and searches extensions to layers $1, \dots, n - 1$ subject to canonical-augmentation constraints (stabilizer-symmetric candidates are pruned unless they lie in the group generated by the layer-0 stabilizer).

2.4 Forbidden masks and the admissibility prune

For each unordered 4-subset of already-placed points, the fifth point completing a degenerate 5-set is forbidden in each remaining layer; the sphere table precomputes, per 4-subset of G_n , the bitmask of forbidden positions per layer ($\binom{125}{4} \cdot 5$ bytes ≈ 0.5 GB at $n = 5$; $\binom{216}{4} \cdot 6$ bytes ≈ 4.2 GB at $n = 6$). The depth-first search maintains, per future layer, the union mask of forbidden positions, and consults it in three ways: singleton blocking, pair and triple masking against candidate layer sets (precomputed compatibility masks), and an *admissibility bound* — for each future layer k , a lower bound on the size k must contribute, obtained by subtracting from the target the currently placed count and the maximal contribution of the other unplaced layers.

2.5 The corrected prune

The first implementation of the admissibility bound used, at depth j and for a future layer $k > j$, the quantity $k - m - 4(4 - k)^1$ as the lower bound on $|S_k|$. This counts only the layers *after* k as slack, ignoring the still-unplaced layers $j + 1, \dots, k - 1$, and therefore over-estimates the requirement on layer k — an unsound prune. The corrected bound, $low = K - m - 4(NZ - 1 - j)$, allocates slack over all unplaced layers other than k and is the version used in the complete run. The correction changed the search tree size by roughly a factor of two (591M vs. 292M nodes), and the earlier, incomplete run’s conclusion is superseded by the corrected one. We

¹Concretely, the code computed $low = K - m - 4(NZ - 1 - k)$ with $NZ = 5$, an over-estimate obtained by counting only layers strictly after k instead of all unplaced layers other than k .

record this explicitly because it is exactly the class of error — a sound-looking prune that silently discards live branches — that an incomplete enumeration can hide; the positive control (Section 4) is the safeguard that caught it.

2.6 Incremental mask propagation

The dominant cost of the table-driven search is rebuilding the per-layer forbidden masks at each node: a naive implementation performs $\binom{m}{4}$ random lookups into the sphere table per node. The optimized enumerator propagates the masks incrementally: when a layer set C is placed, only the 4-subsets meeting C change the masks, and the 1-of- C terms are accumulated for free during the per-position compatibility computation already needed for candidate generation. The rewrite was validated by node-count equivalence: on 30 consecutive orbit representatives the incremental binary produced exactly the node counts of the original implementation (e.g. 3,620,318 nodes on representative 0 in both), at roughly $6\times$ throughput.

3 Results

n	$C(n)$	evidence	fits $\lfloor (5n+7)/2 \rfloor$?	record
3	8	exhaustive	8	first exact
4	11	exhaustive	11	first exact
5	14	complete layer enumeration	14	first exact
6	≥ 18	verified certificate; $k = 19$ decision in progress	18	pending
7–12	$\geq$ record	public certificates	exact match of the law	public record

Table 1: Exact values and bounds for $C(n)$.

4 Verification

Three independent checks guard the claims. (i) *Table self-test*: the sphere table is verified entrywise against direct evaluation (0 mismatches). (ii) *Positive control*: the same binary run at $k = 14$ finds a 14-subset; the found set is independently verified by the lifted-determinant checker (all $\binom{14}{5} = 2002$ determinants nonzero, minimum absolute value recorded). (iii) *Candidate cross-check*: the layer-candidate count 11,824 is reproduced by an independent brute-force enumeration of layer 4-subsets. The $n = 5$ run completed all 1,905 orbits with no deadline truncations. A cube-and-conquer LRAT witness for the $n = 5$, $k = 15$ instance is an optional second certificate, currently incomplete on its splitting frontier; the enumeration stands as the decision.

4.1 Status of the $n = 6$ decision

At time of writing the $k = 19$ enumeration on G_6 is in progress (canonically ordered representatives, incremental-mask binary, 16 worker threads). A completed run reporting *found* = 0 over all 8,133 representatives would establish $C(6) = 18$; a found configuration would raise the lower bound to 19. The run is checkpointed per orbit, and its logs will accompany the released artifacts. (This section will be finalized when the run completes.)

5 Conclusion

The first exact values of the no-five-on-a-sphere function confirm the empirically fitted law $\lfloor (5n+7)/2 \rfloor$ at every n where it has been decided, and the method — canonical layer enumeration over a tabled degeneracy oracle — is strong enough to decide $n = 6$ at moderate cost and to provide, at $n = 5$, the first optimality proof in this family. The same machinery extends in principle to $n = 7$, though the table size ($\binom{343}{4}$ sphere rows) currently exceeds workstation memory and would require an out-of-core or compressed representation. Separately, the corrected-prune episode (Section 2.5) is a reminder that exhaustive-search claims earn their certainty from controls, not from the absence of counterexamples in a partial log.

References

- [1] Google DeepMind, *AlphaEvolve: a repository of problems*, problem 60 (no five points on a sphere or plane), 2025.
- [2] Demonstrandum Research, artifact bundle of certificates for the no-five-on-a-sphere problem, June 2026; fitted law $C(n) = \lfloor (5n + 7)/2 \rfloor$.
- [3] milesandmistakes, certificate repository for AlphaEvolve problem 60, August 2026.
- [4] Numaro, search report for AlphaEvolve problem 60, $n = 14$ –26, July 2026 (no coordinates published).